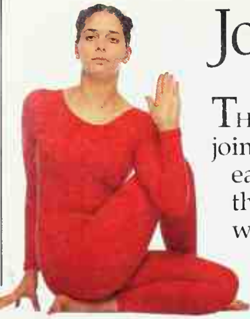
Joints between bones

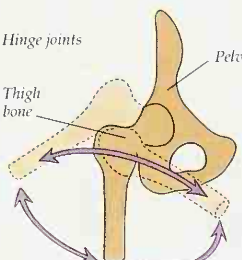


THE 206 SKELETAL BONES ARE LINKED together at joints. Ancient people probably first noticed these when eating freshly killed meat, as they tried to dismember the animal by cutting off a limb. They found that while the limb moved to and fro easily, it was extremely difficult to tear away from the body or to cut right through the joint itself. Since that time, joints have often been studied after death, but rarely in the active, living body. Recently, scanning techniques such as CAT (Computerized Axial Tomography) and MRI (Magnetic Resonance Imaging) reveal how a joint such as the knee or shoulder can withstand tremendous forces and stresses.

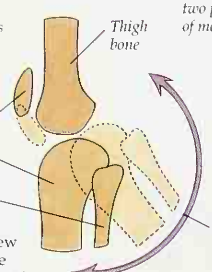
SUPPLE JOINTS
Like any body part, joints benefit from use and deteriorate with neglect. Activities such as yoga promote the full range of joint movement, encourage maximum flexibility, and help to postpone the stiffness, pain, or discomfort that sometimes arrive with the onset of old age.

JOINTS GALORE
The hand is an astonishing example of muscle versatility, pulling with precision on 19 movable joints – not counting those in the wrist. The first knuckle joint at the base of each finger and thumb needs good mobility and has a ball-and-socket design. All the other knuckles are hinge joints, allowing the fingers to bend over toward the palm, but not toward each other. Between the carpals (p. 15) are "gliding" joints, so that the wrist bones can slide against each other easily.

BALLS AND HINGES
The body's joints may have inspired mechanical equivalents invented by engineers. The hip and shoulder are ball-and-socket joints. The rounded end of the thigh or upper-arm bone swivels in a cup-shaped socket in the hip bone or shoulder blade. Like a computer joystick joint, this design allows movement in two planes: front to back and side to side. The knee and elbow are hinge joints, as on a door. They have a more limited movement, mainly in one front-to-back plane.



Front view of ball-and-socket joint (hip)



Side view of hinge joint (knee)